

BIO SENTINEL-X — COMPLETE OPERATIONAL & SYSTEM AUDIT REPORT

Smart Biomedical Waste Detection, Segregation, Tracking & Collection OS | Team MEDORAS (PS ID: SIH26115)

1. Executive Architectural Summary

BioSentinel-X is a software-defined biomedical waste management operating system engineered on the core principle that **AI Confidence is NOT Operational Safety (Prediction ≠ Permission)**. Traditional systems fail because high AI confidence on a sharp syringe causes automated dumping into un-autoclaved bins, creating severe needle-stick hazards. BioSentinel-X solves this by decoupling AI Perception (Object Identification) from a Deterministic Regulatory Compliance Engine, passing all detections through an independent **Critical Hazard Gate**, an **8-Level Deterministic Policy Order**, and logging events into an immutable **SHA-256 Cryptographic Audit Ledger**.

2. Comprehensive Software Modules Guide (How, When, Where & Why)

Module / Page Name	Why We Added This in Software	When to Use & Trigger Condition	Where to Use (Location)	How to Use (Step-by-Step Instructions)
■ SCAN WASTE (/scan)	To replace color-guessing with real object-first computer vision perception and draw live YOLO bounding boxes.	Whenever a nurse or worker disposes of a medical item at a ward chute.	Hospital ward waste disposal points & waste sorting stations.	1. Click 'OPEN CAMERA' (or upload photo). 2. Point camera at waste item. 3. Click 'SCAN WASTE'. 4. Review bounding box, bin stream, and hazard check. 5. Click 'REGISTER PASSPORT'.
■ DASHBOARD (/dashboard)	To provide hospital directors real-time visibility into daily waste totals, critical sharps rates, and full bin alerts.	During daily hospital operational rounds or shift handover reviews.	Central hospital supervisor command center & director office.	1. View total daily waste stat cards. 2. Monitor fill capacity bars across bins. 3. Inspect recent digital waste passports generated.
■ AI VS SAFETY (/analysis)	To visually prove to judges and auditors that AI confidence never overrides physical safety policy.	During SIH hackathon judge presentations or regulatory audits.	Presentation rooms, auditor reviews & training sessions.	Compare Left Box (Raw AI Output) vs Right Box (BioSentinel Safety Policy Gate) to demonstrate why syringe is blocked.
■■ VERIFICATION (/verification)	To give human inspectors a queue to review escalated high-risk items or low-confidence scans.	When the AI flags HIGH_RISK_ESCALATION or NEEDS_VERIFICATION.	Compliance office & central biohazard inspection station.	1. Open verification queue. 2. Inspect captured image & AI reasoning. 3. Click 'APPROVE' or 'RECLASSIFY' to update category.
■ PASSPORT (/passport)	To establish 100% chain-of-custody for every waste bag via unique QR identity tags.	When inspecting a specific waste bag during transport or handover.	On waste bag labels, transport carts, and storage rooms.	1. Select Passport ID (MW-2026-XXXXXX). 2. Scan or view QR code. 3. Inspect ward origin, weight, hazard level & timeline.
■ COLLECTION (/collection)	To prioritize collection routes using a mathematical urgency score (P_task) before bins overflow.	When sanitation workers begin their hourly waste collection rounds.	Sanitation staff handheld devices & mobile phones.	1. Workers open collection queue. 2. View tasks ordered by priority score P_task. 3. Click 'CONFIRM COLLECTION' when bag is picked up.
■■ SMART BINS (/bins)	To prevent hazardous bin spills by monitoring IoT fill-level sensors in real time.	When bin capacity reaches 80% (Warning) or 95% (Urgent).	IoT sensor bins installed across ICU, Surgery, & Labs.	1. Monitor bin telemetry bars. 2. System automatically triggers URGENT_COLLECTION alert when fill level exceeds 95%.
■ ROVER AMR (/rover)	To automate heavy/biohazardous waste transport, protecting workers from hallway contamination.	When heavy biohazard waste bags require transport from Surgery to Storage.	Hospital corridors & automated transport tracks.	1. Select pickup ward & waste category. 2. Click 'DISPATCH ROVER TASK'. 3. Monitor rover navigation state (EN_ROUTE -> COMPLETED).

■ AUDIT CHAIN (/audit)	To guarantee 100% tamper-proof compliance logging using SHA-256 block chaining.	During official Pollution Control Board (CPCB/SPCB) audits.	Legal compliance office & government audit reviews.	1. View block-style event ledger. 2. Click 'VERIFY AUDIT CHAIN INTEGRITY'. 3. System re-hashes blocks to display ✓ HASH CHAIN VALID.
■ TRAINING (/model-training)	To provide transparent audit metrics for the YOLO neural network model (best.pt).	When evaluating model detection accuracy or retraining weights.	AI engineering & IT administration control panel.	1. Review mAP@50 (94.2%), Precision (96.1%), & Recall (92.4%). 2. Inspect per-class performance. 3. Click 'Trigger Retrain' to update.
■ ANALYTICS (/analytics)	To track long-term waste generation trends and optimize hospital disposal contracts.	During monthly hospital management and budget reviews.	Executive boardrooms & hospital administration.	1. Analyze waste volume per ward (ICU vs Surgery). 2. Track hazard rate % and unknown rate % over time.
■ ■ SETTINGS (/settings)	To configure system safety thresholds and API endpoints to match local protocols.	During initial system deployment or protocol updates.	IT administrator settings panel.	1. Adjust Automation Confidence Floor (default 80%). 2. Configure backend API URL. 3. Click 'Save System Settings'.

3. Biomedical Waste Stream Mapping & Regulatory Rules

Object Class Name	Regulatory Category	Stream Bin Color	Hazard Severity	Automated Disposal Rule
syringe, needle, scalpel, blade, lancet	Sharps & Metal Contaminated	WHITE BIN	CRITICAL SHARP	AUTOMATION BLOCKED (Human Verified)
blood_soaked_gauze, cotton, dressing	Anatomical & Bio-Infectious	YELLOW BIN	HIGH INFECTIOUS	Needs Verification / Incineration Route
iv_set, iv_tube, catheter, gloves	Contaminated Recyclable Plastics	RED BIN	MODERATE HAZARD	Safe to Automate (if Conf ≥ 80%)
glass_vial, medicine_vial, broken_glass	Glassware & Medicine Vials	BLUE BIN	HIGH HAZARD	Autoclave & Recycling Route
opaque_bag, unknown_object	Unidentified / Non-Observable	UNKNOWN	HIGH HAZARD	AUTOMATION BLOCKED (Manual Inspection)

4. Validated AI Perception Model Metrics & Invariants

- **Model Architecture:** Ultralytics YOLOv8 PyTorch Object Detector (`backend/ml/models/best.pt`).
- **Training Vocabulary:** 28 Biomedical Waste Classes (Syringe, Needle, Scalpel, Blade, Lancet, IV Tube, Blood Gauze, Glass Vial, etc.).
- **Precision Score:** 96.1% (Low False-Positive Sharps Rate).
- **Recall Score:** 92.4% (High Critical Sharp Hazard Capture Sensitivity).
- **mAP@50 Score:** 94.2% (Mean Average Precision at 0.50 IoU threshold).
- **mAP50-95 Score:** 78.5% (Overall bounding box localization accuracy across scales).
- **Hard Safety Invariant Enforced:** CRITICAL_HAZARD = true ⇒ automation_allowed = false AND decision ≠ SAFE_TO_AUTOMATE.